FOR A NOETHERIAN RING THE COMPLETION AND LOCALIZATION ALONG AN ELEMENT EXHIBIT FINITELY GENERATED MODULES AS A FIBRE PRODUCT

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement span contains the author's displayed fibre product of module categories, delivered verbatim. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Soundness of the completion in the Noetherian case: flatness of R -> R-hat and M-hat = M tensor_R R-hat for finite M. used at proof lines 25015-25020. Identifies the displayed functor with the one of the formal gluing theorem.
- The formal gluing theorem as an equivalence of unrestricted module categories. invoked at proof lines 25022-25027. Supplies full faithfulness and the existence of the glued module M.
- Faithful flatness of R -> R-hat x R_f and the cover criterion for finite generation. used at proof lines 25028-25031. Upgrades the glued module to a finitely generated one.

Source reference. Original lemma 00MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 817–817. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00MA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 816–816. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00EO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 542–542. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original theorem 24962: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24962–24962. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Proposition 0.2. *Let R be a Noetherian ring. Let $f \in R$ be an element. Let $R^\wedge$ be the f-adic completion of R. Then the functor $M \mapsto (M^\wedge, M_f, can)$ defines an equivalence*

$$Mod_R^{fg} \longrightarrow Mod_{R^\wedge}^{fg} \times_{Mod_{(R^\wedge)_f}^{fg}} Mod_{R_f}^{fg}$$

Proof. The ring map $R \to R^\wedge$ is flat by Algebra, Lemma 00MB. It is clear that $R/fR = R^\wedge/fR^\wedge$. By Algebra, Lemma 00MA the completion of a finite R-module M is equal to $M \otimes_R R^\wedge$. Hence the displayed functor of the proposition is equal to the functor occurring in Theorem 24962. In particular it is fully faithful. Let (M_1, M_2, ψ) be an object of the right hand side. By Theorem 24962 there exists an R-module M such that $M_1 = M \otimes_R R^\wedge$ and $M_2 = M_f$. As $R \to R^\wedge \times R_f$ is faithfully flat we conclude from Algebra, Lemma 00EO that M is finitely generated, i.e., $M \in \mathrm{Mod}_R^{fg}$. This proves the proposition. □